STA 712 Exam 1 Review

Information: The midterm exam will cover class material through October 7, and homeworks 1 – 6. The questions below are not completely comprehensive, but will give you a sense of the kinds of questions I could ask on the exam.

EDM Questions

1. Suppose we are told that $Y \sim EDM(\mu, \phi)$, and we know that $V(\mu) = \mu + \dfrac{\mu^2}{r}$, where r is a **known** constant.

   (a) Find the canonical parameter θ as a function of μ. *Hint*: $\dfrac{1}{\mu + \mu^2/r} = \dfrac{1}{\mu} - \dfrac{1}{\mu + r}$

   (b) Find the cumulant function $\kappa(\theta)$ as a function of θ.

   These values of θ and $\kappa(\theta)$ match the values for a negative binomial distribution, so if $Y \sim EDM(\mu, \phi)$ and $V(\mu) = \mu + \dfrac{\mu^2}{r}$, then Y must be negative binomial!

2. Determine which of these functions are suitable link functions for a GLM. For those that are not suitable, explain why not.

   (a) $g(\mu) = e^{\mu}$ when $-\infty < \mu < \infty$
   (b) $g(\mu) = e^{-\mu}$ when $-\infty < \mu < \infty$
   (c) $g(\mu) = \log\left(\dfrac{\mu}{\mu + 1}\right)$ when $\mu > 0$
   (d) $g(\mu) = \log(\mu)$ when $-\infty < \mu < \infty$

3. Determine which of the following distributions are EDMs by identifying (where possible) $\theta, \kappa(\theta)$, and ϕ.

   (a) Beta distribution:
   $$f(y; a, b) = \frac{\Gamma(a+b)}{\Gamma(a)\Gamma(b)} y^{a-1}(1-y)^{b-1}$$
   for $0 < y < 1$, $a > 0$, and $b > 0$.

   (b) Strict arcsine distribution:
   $$f(y; p) = \frac{A(y)}{y!} \exp\{y \log(p) - \arcsin(p)\}$$
   where $y = 0, 1, 2, ...$ and $0 < p < 1$, and $A(y)$ is a complicated function depending only on y.

4. (a) Let $Y \sim Geometric(p)$, with probability function
   $$f(y; p) = p(1-p)^y$$
   for $y = 0, 1, 2, ...$ and $0 < p < 1$. (This parameterization of the geometric distribution counts the number of failures until the first success, where p is the probability of success). Show that the geometric distribution is an EDM by identifying θ, $\kappa(\theta)$, and ϕ.

(b) Use part (a) and properties of EDMs to find $\mathbb{E}[Y]$ and $Var(Y)$.

(c) Find the unit deviance $d(y, \mu)$.

(d) Suppose that we observed iid sample $(\mathbf{x}_1, Y_1), ..., (\mathbf{x}_n, Y_n)$ from some distribution, where

$$Y_i|\mathbf{x}_i \sim Geometric(p_i)$$
$$h(p_i) = \mathbf{x}_i^T \boldsymbol{\beta}$$

If $h(p_i) = g(\mu_i) = \theta_i$, write down either $h(p_i)$ or $g(\mu_i)$.

(e) For the model in part (d) (using the canonical link), write down an expression for $U(\boldsymbol{\beta})$ and $\mathbf{H}(\boldsymbol{\beta})$. You do **not** need to re-derive these from the log-likelihood – you may rely on general EDM results from class.

5. Recall that the **cumulant generating function** $C_Y(t)$ for a random variable Y is defined by

$$C_Y(t) = \log \mathbb{E}[e^{tY}]$$

(This definition is not specific to the EDM family). Suppose we have two independent random variables Y_1 and Y_2 (not necessarily EDMs), for both of which the cgf exists at t. Show that

$$C_{Y_1+Y_2}(t) = C_{Y_1}(t) + C_{Y_2}(t)$$

6. An important property of EDMs (as we have defined them) is that they are **reproductive**. If $Y_1, ..., Y_n \overset{iid}{\sim} EDM(\mu, \phi)$ then

$$\overline{Y} = \frac{1}{n}\sum_{i=1}^{n} Y_i \sim EDM\left(\mu, \frac{1}{n}\phi\right)$$

(a) In class, we showed that the cgf of $Y \sim EDM(\mu, \phi)$ is

$$C_Y(t) = \frac{\kappa(\theta + \phi t) - \kappa(\theta)}{\phi}$$

Use this result to show that $\overline{Y} \sim EDM\left(\mu, \frac{1}{n}\phi\right)$.

(b) Using part (a), argue that – for small values of ϕ – if $Y \sim EDM(\mu, \phi)$ then $Y \approx N(\mu, \phi V(\mu))$.

Logistic regression questions

7. Let Y_i^* be a *continuous* random variable, defined by

$$Y_i^* = \mathbf{x}_i^T \boldsymbol{\beta}^* + \varepsilon_i$$

where the ε_i are iid samples from the **Gumbel distribution**, with pdf

$$f_\varepsilon(t) = \exp\{-(t - e^{-t})\} \quad t \in (-\infty, \infty)$$

Then, let $Y_i = \mathbb{I}\{Y_i^* > 0\}$. Show that $Y_i|\mathbf{x}_i \sim Bernoulli(p_i)$, with

$$\log(-\log(1 - p_i)) = \mathbf{x}_i^T \boldsymbol{\beta}$$

(where $\boldsymbol{\beta} = -\boldsymbol{\beta}^*$).

The function $\log(-\log(1 - p_i))$ is called the **complementary log-log** (cloglog) link.

8. Your friend fits a logistic regression model, and wants to check the quality of the predicted probabilities. They notice that $\frac{1}{n}\sum_{i=1}^{n}\widehat{p}_i = \frac{1}{n}\sum_{i=1}^{n}Y_i$ for their fitted model, and they claim this is evidence that the model is well-calibrated. Comment on your friend's interpretation.